

XAI Compensation Equity: Compliance Audit Report

Section 1: Executive Summary

This report details the pay equity audit findings across the organisation. The Total Unadjusted Pay Gap is 32.1%. Of this, the decomposition isolated an Unexplained Penalty of 27.3% applied persistently to female employees, translating to an average penalty of ■294,489 per employee.

Note: this synthetic dataset was designed to isolate the direct compensation penalty independent of structural concentration effects. In real organisations, the explained component is typically 40-65% of the total gap (Keka India 2025-26). A low explained component in this audit indicates the simulated bias is primarily direct rather than structural.

Section 2: Regulatory Status

RED FLAG: Unexplained pay gap of 27.3% exceeds the threshold for potential ER Act liability. Estimated affected employees: 2320. Estimated annual remediation cost: ■849,509,375 - ■993,742,000 at 95% confidence.

The Equal Remuneration Act (1976) mandates identical remuneration for identical work. An isolated direct bias exceeding statistical tolerance represents a critical compliance failure requiring immediate remediation.

Section 3: Primary Bias Vectors

Manager evaluation scores show a -0.261 mean difference between male and female employees, accounting for approximately ■40,567,295 of the explained pay gap annually. Subjective manager assessment is the primary legitimate pathway through which compensation disparity operates in this organisation.

Section 4: Quantile Coverage Reliability

The underlying LightGBM regression model successfully satisfied the P05 to P95 empirical band coverage constraints (>80%), guaranteeing high statistical confidence in the baseline median (P50) predictions utilized by the SHAP explainer to calculate the above residuals.

Section 5: Remediation Strategy

An affected employee is defined as any female employee whose predicted P50 salary is more than one standard deviation below the male median for their specific job level.

Based on this strict criteria, a budgeted remediation pool of ■849,509,375 – ■993,742,000 at **95% confidence** must be provisioned to surgically adjust the base CTC of the 2320 heavily-underpaid female personnel without inducing reverse-bias.

Methodological Context: This remediation budget representing approximately 7.7% of total payroll is intentional. While real-world equity corrections typically cost 0.5%–2.0% of payroll, this 7.7% figure is the precise mathematical consequence of our synthetic dataset design—which intentionally injected an extreme 28.4% direct pay penalty across 40% of the workforce to stress-test the XAI detection bounds.